

Lecture 26

3/20/26

JRA

Contrast (revisited)

Strength of echo is source of contrast in US

Three types of reflectors in body

(1) Specular - mirror-like reflector well-defined reflection direction

Can calculate reflection

Diaphragm

See bb

Z is called acoustic impedance

$$R = \frac{I_r}{I_t} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

analog of refractive index

Compared formula w/ Fresnel Eq for light

Very impt for sound waves

Z s very similar for soft tissue - see PP table

Interface between soft tissues $\Rightarrow \approx 1\%$ reflection

Transmission impt for visualizing

(non-smooth)
(2) Diffuse Reflectors



Rough surface
→ normal varies
→ reflection direction varies

Transducer only picks up fraction of reflection (the part going toward it)

Difficult to quantify but doctor knows what normal tissue looks like

(3) Scattering by RBCs that gives blood signal
Multi direction

Implications

- (a) Soft tissues have similar "z" values
⇒ $R \leq 1\%$ and most US transmitted, allowing imaging of deeper structures
- (b) Air/Soft tissue interfaces

$R \approx 1$ or 100% ⇒ No transmission

⇒ Doctor uses gel to match impedances

Air has super small Z
ST relatively large Z

Make Z_s more similar

Decreases R

Get transmission into body

Looked at Ranger picture

- (c) Reflections at bone/ST interfaces are big
 → get surface image of bone
 because no transmission to give subsequent ^{deeper} reflections

SIMULATION

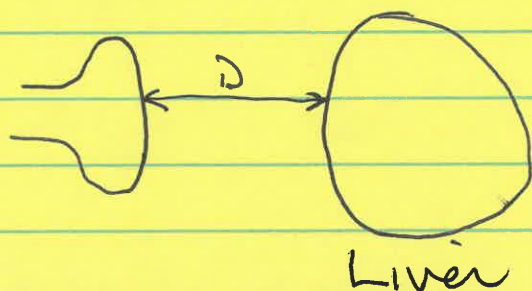
Full sim - 2 sides

No transmission - 1 side

Time scale for image ≈ 2 msec

ACTIVITY 9

13- Microsecond Rule for
 Calculating Tissue Depth



Unaccelerated motion

$$D = v t$$

t_{echo} = go-return time

$$2D = 1540 \text{ m/sec} \cdot t_{\text{echo}}$$

$$D = \frac{1540 \text{ m/sec}}{2} \cdot t_{\text{echo}}$$

Punch Line: Every additional 1 cm of depth corresponds to 13 μsec of go-return time

Insert $D = 1 \text{ cm}$ should

(A) Proof

(B) Lens Front 6 μsec
 Lens back 12 μsec

$$\begin{aligned} \Delta t &= 6 \mu\text{sec} \\ &\approx \frac{1}{2} 13 \mu\text{sec} \\ &\Rightarrow \frac{1}{2} \text{ cm thick lens} \end{aligned}$$

$$(A) \quad D = \frac{1540 \cdot t_{\text{echo}} \text{ m}}{2} \quad 10^{-2} \text{ m}$$

$$t_{\text{echo}} = \frac{2 \cdot 10^{-2}}{1540} \text{ sec}$$

$$= 0.00012987 \approx 13 \mu\text{sec}$$

(B) 1 cm 13 μsec
 $\frac{1}{2}$ cm 6.5 μsec

Difference $\approx 6 \mu\text{sec}$
 for lens so lens
 $\approx 0.5 \text{ cm thick}$

SIMULATION

Fetal Heart rate

Typically $\approx 150-180$ beats/minImaging Modalities

There are 3 common imaging modalities

- 1) A Mode ^{amplitude} 1D imaging Used to measure eye dimensions
ophthamology
- 2) M Mode ^{motion} 1D imaging spikes as a fcn of time
- 3) B Mode ^{brightness} 2D imaging slices
workhorse of modern US that look like a "traditional image"

M mode is popular in cardiology when you want to go fast
Measure fetal heart rate

Brightness - grayscale is used to encode echo strength

B-mode is fast enough to do real-time imaging (even if slower than M mode)
 ~ 30 frames/sec

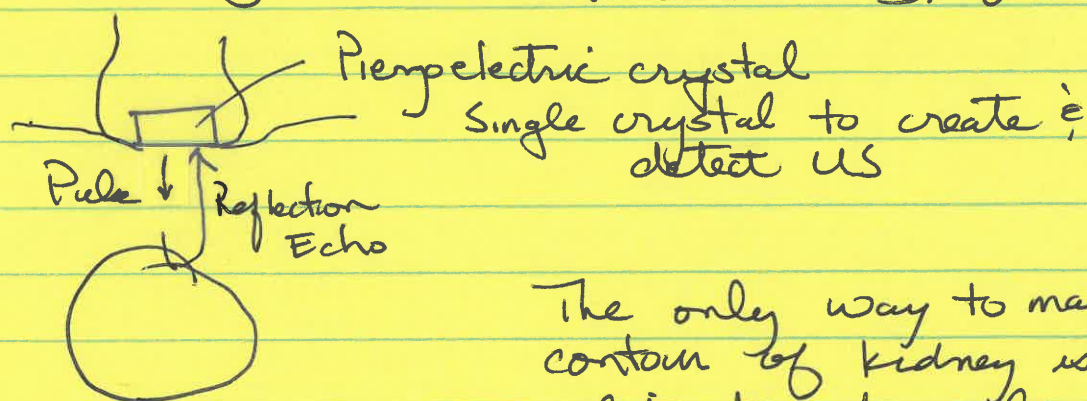
26-6

How do you get 2D image?
from a transducer ^{source & detector}

A & M modes were only modes available for a long time

Older transducer

1D imaging $\Rightarrow$ 2D imaging w/ moving ^{transducer} by hand



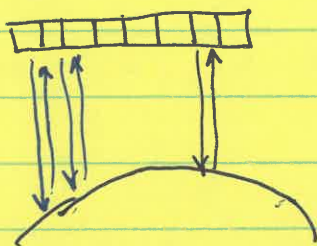
Map shape by moving xtal along skin

The only way to map contour of kidney is to slide transducer along skin

$\Rightarrow$ NOT accurate

Need transducer w/ many xtals

Newer transducers have many xtals (up to 512)



Fire subset at a time (e.g. 20)

Fire ~ 20 xtal's at a time

Proceed sequentially & collect to build up image

Beam can also be electronically steered

Beam Steering next: after break ^{to this in} cardiology

Know how to quantify depth
get contrast

What about resolution
Need resolution

This dictates choice of frequency

2 prominent types of resolution

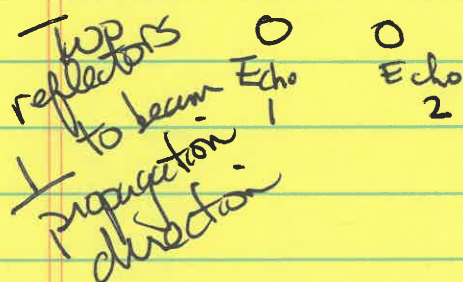
1 more will be suppressed

- Lateral
- Axial

Beam means energy concentrated over finite area

Width
= cont det

Lateral



Question: Collect series of echoes
To resolve, need echo free region between reflectors

Want Echo 1 ^{Echo}Free Echo 2

How do you get this?

Reflectors
separated
along
direction
of beam
propagation



○

Echo 1

○

Echo 2

To resolve need

Echo

Echo Free

Echo

This is what's required
for axial resolution

Pulse length = $\frac{c}{\Delta f}$

Short 1
High 2

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